GAUSSIAN MIXTURE MODELS

Fri | Mon

Agenda

Recap kmeans

Hard clustering

1D | 2D gaussian

GMM

EM

Customer segmentation

Recap → DATA

goal →

kmeans algo

100 - 200

— (1) init

→ we init k centroids

→ kmeans ++

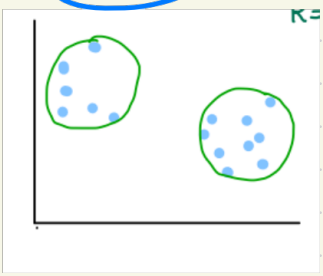
(2) iterate until convergence

Reiterating

(a) assign dp to nearest centroid

(b) update centroids

k=2



VIS^n

TSNE

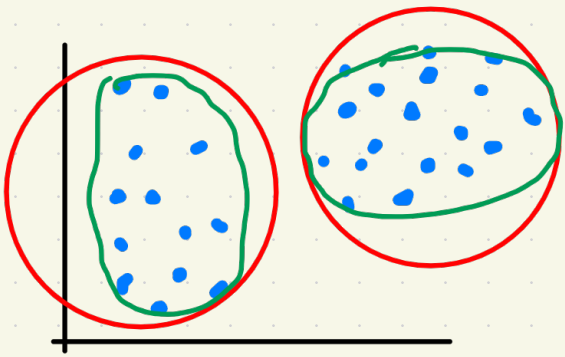
VIZ^n

features → ELBOW

Business

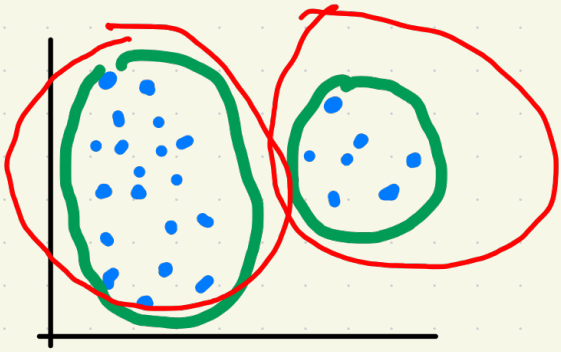
Challenges | limitations → kmeans

1. Globular structure

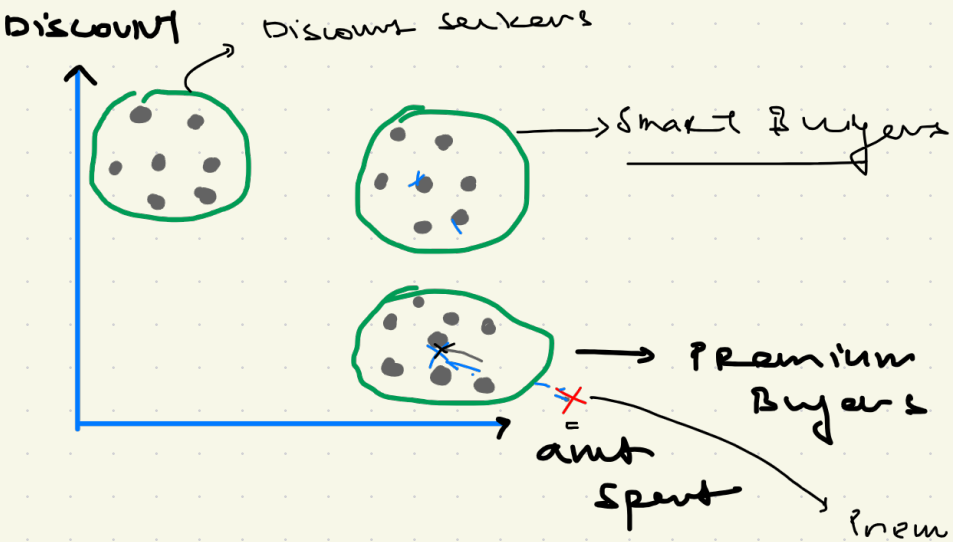


2. Diff. sizes

kmeans

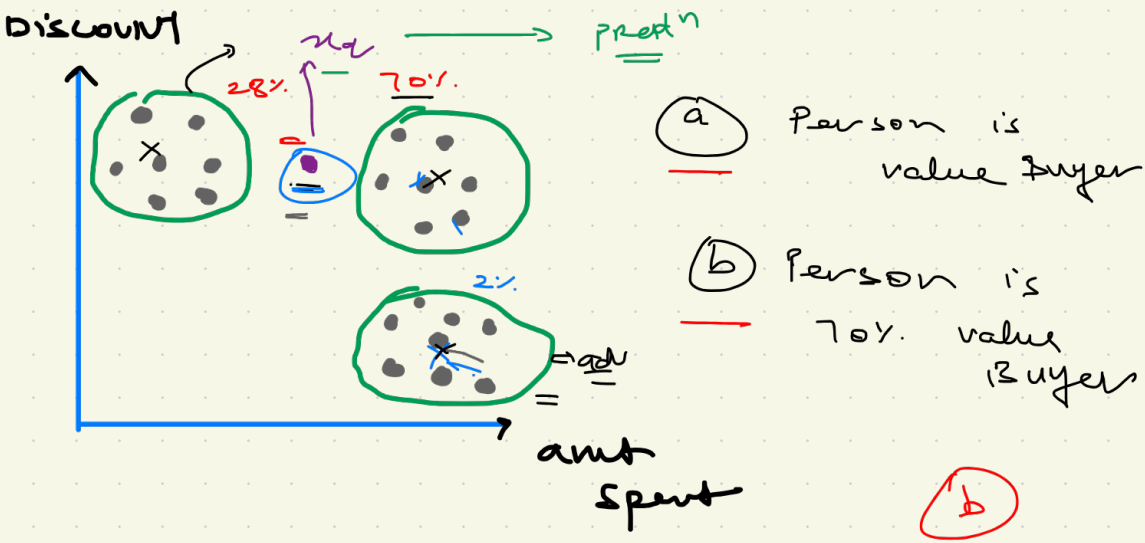


3. Hard clustering



Premium → predict SKlearn

(Prediction in kmeans)



(a) Person is value Buyer

(b) Person is 70% value Buyers

(b) = x

kmeans → Hard clustring → centroid

Soft clustering

Soft clustering ?

kmeans → 3

Amazon → 10$ | customer

Adv

Discount

7$ → Adv

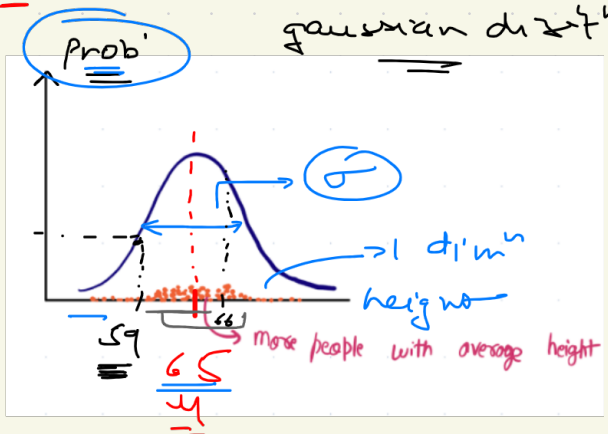
3$ → discounts → Benefits

GMM → will help me do soft clustering

gaussian dist^n

gaussian Dist^n

* heights → PDF



Param → μ, σ

$$P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x_i - \mu)^2}{2\sigma^2}\right)$$

Prob

Area = 1

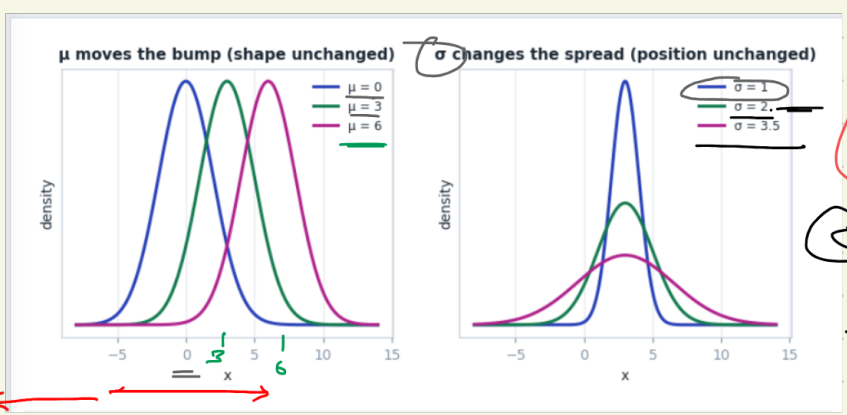
$N(65, 5)$

μ, σ

μ → pos^n / loc^n

↑ μ

↓ μ



↑ σ

↓ σ → spreadness

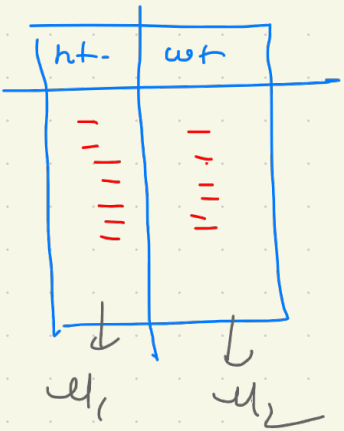
Params → μ, σ → shape of gaussians

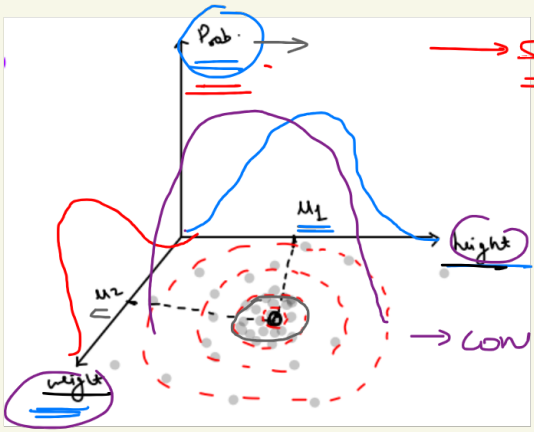
1 dimⁿ

2 dimⁿ

ht + wt.

(2D)





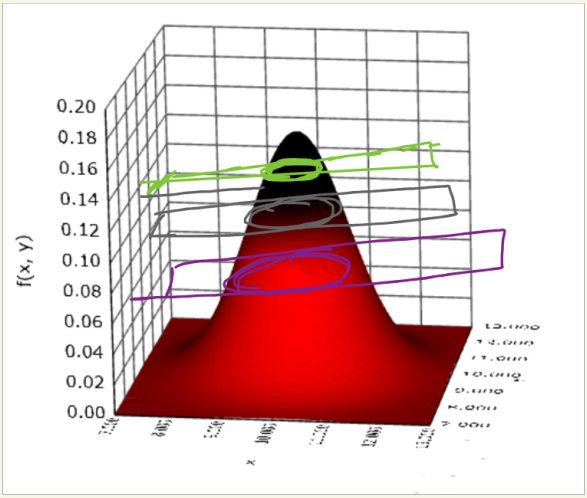
2 dimⁿ

μ → loc^n & shape

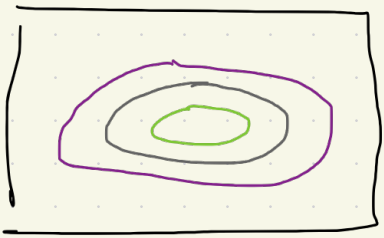
σ

$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$

σ_1, σ_2

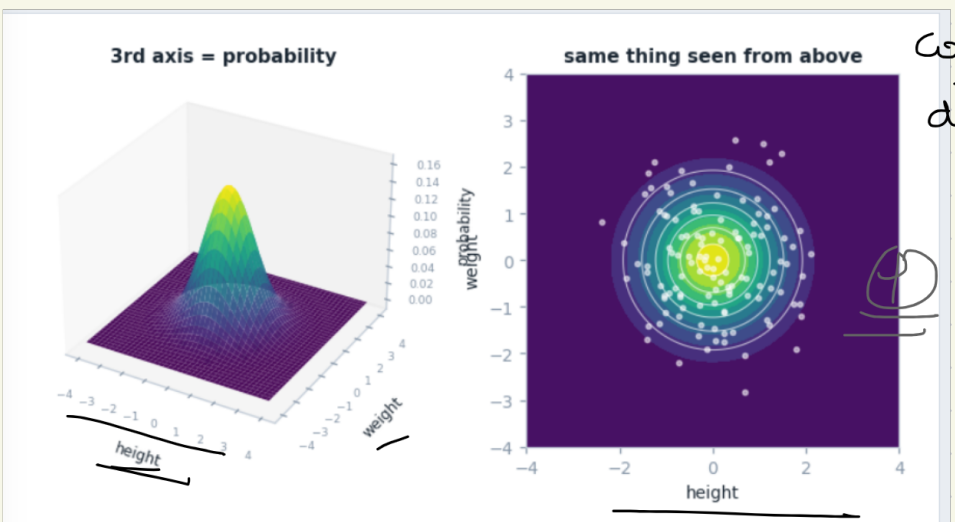


knife



CONTOUR plot

3D



contour

density

covariance

1 Dimⁿ → 2

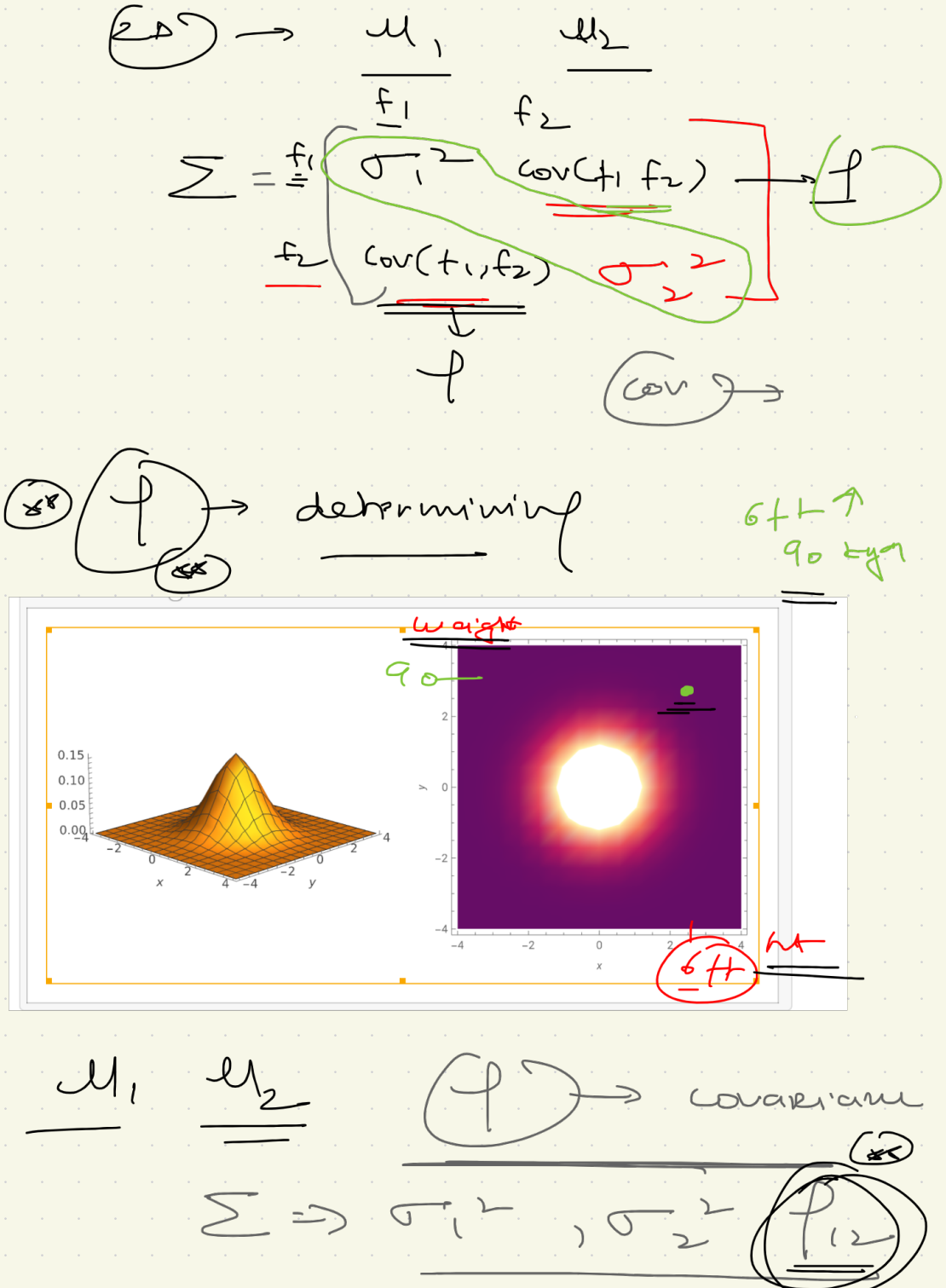
ht wt

2 dimⁿ → μ_1 μ_2 ✓ σ_1 σ_2

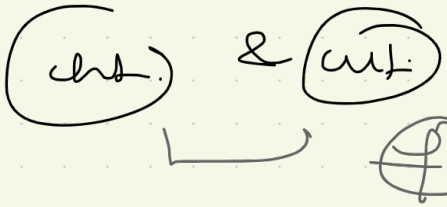
f_1 f_1 f_1 σ_1^2

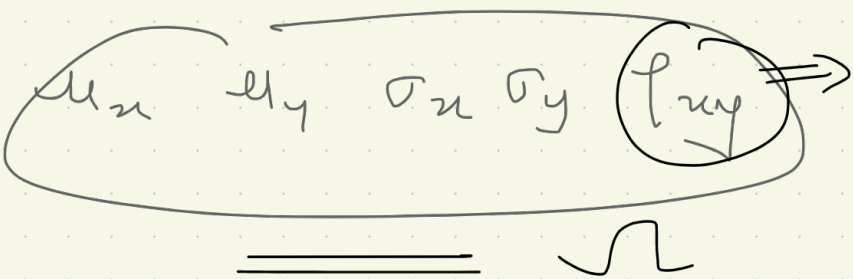
ρ → RHO

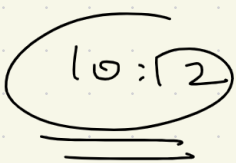
↳ covariance


μ1
μ2
f1
f2
Σ =
σ1²
cov(f1 f2)
cov(f1,f2)
σ2²
cov
determining
6ft
90 kgs
weight
90
0.15
0.10
0.05
0.00
x
y
6ft
μ1
μ2
covariance
Σ ⇒ σ1² , σ2²

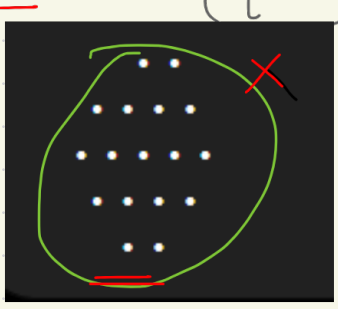




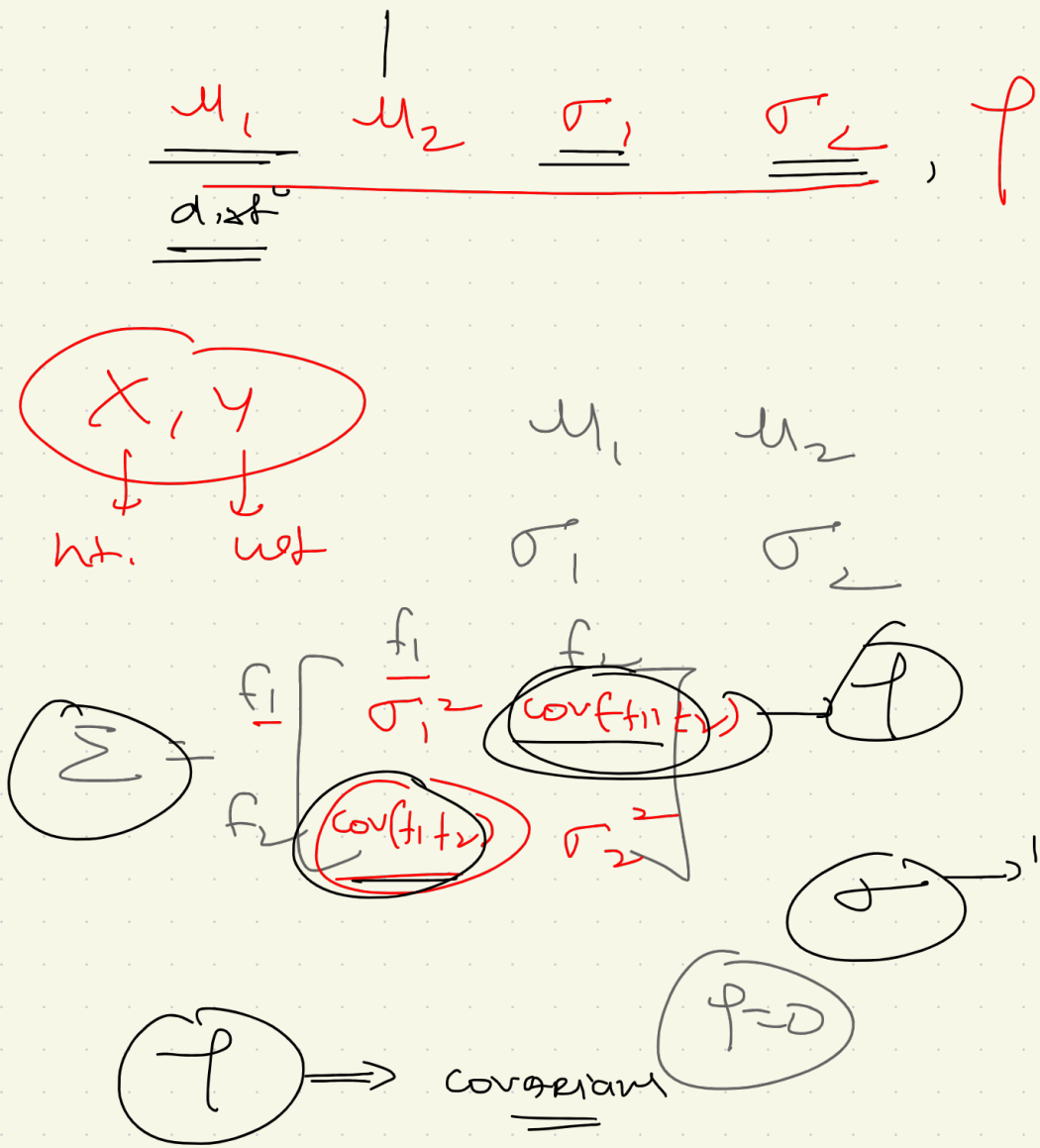




Person	Height	Weight
A	160	50
B	165	55
C	170	60

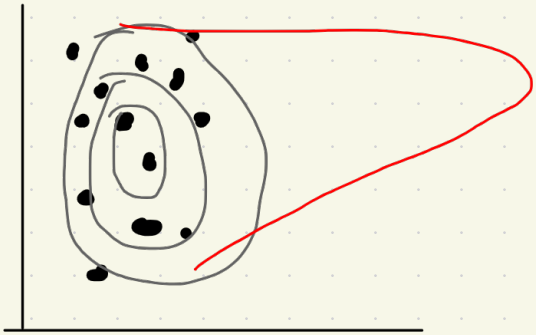






$\mu_x \quad \mu_y \quad \sigma_x \quad \sigma_y \quad \rho_{xy}$







2 dimⁿ

μ_1 μ_2 σ_1^2 σ_2^2

Γ_{12}

2D

3 dimⁿ $\mu = [\mu_1 \quad \mu_2 \quad \mu_3]^T$

	f_1	f_2	f_3
f_1	σ_1^2	=	—
f_2	=	σ_2^2	
f_3	=	—	σ_3^2

3d.

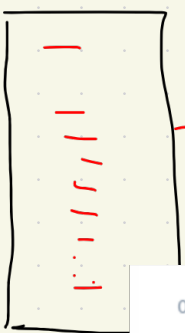
GMM intuition

(K) gaussian mixture models

↓

multiple

hair length (1D)



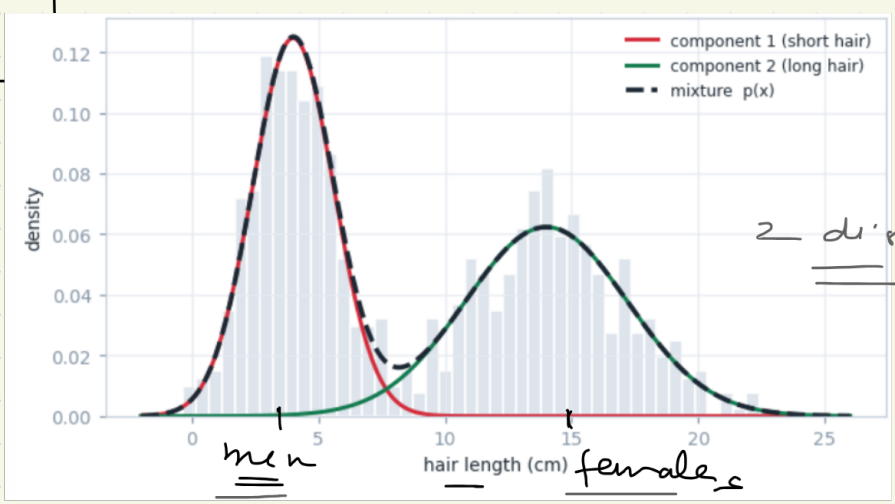
PDF

Hair length

1 peak

⇒

1 distⁿ



2 distⁿ

men

female

Bimodal

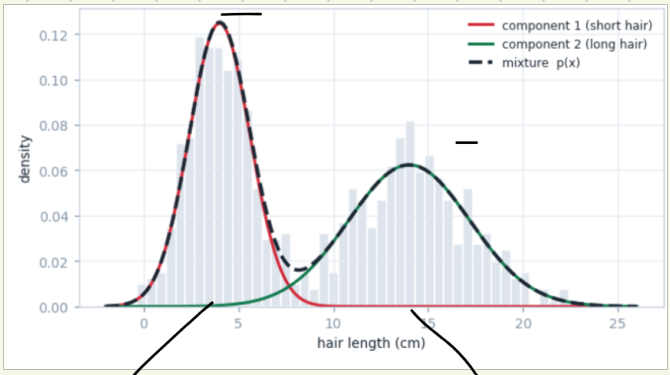
2 distⁿ → 2 clusters

2 men

1 women

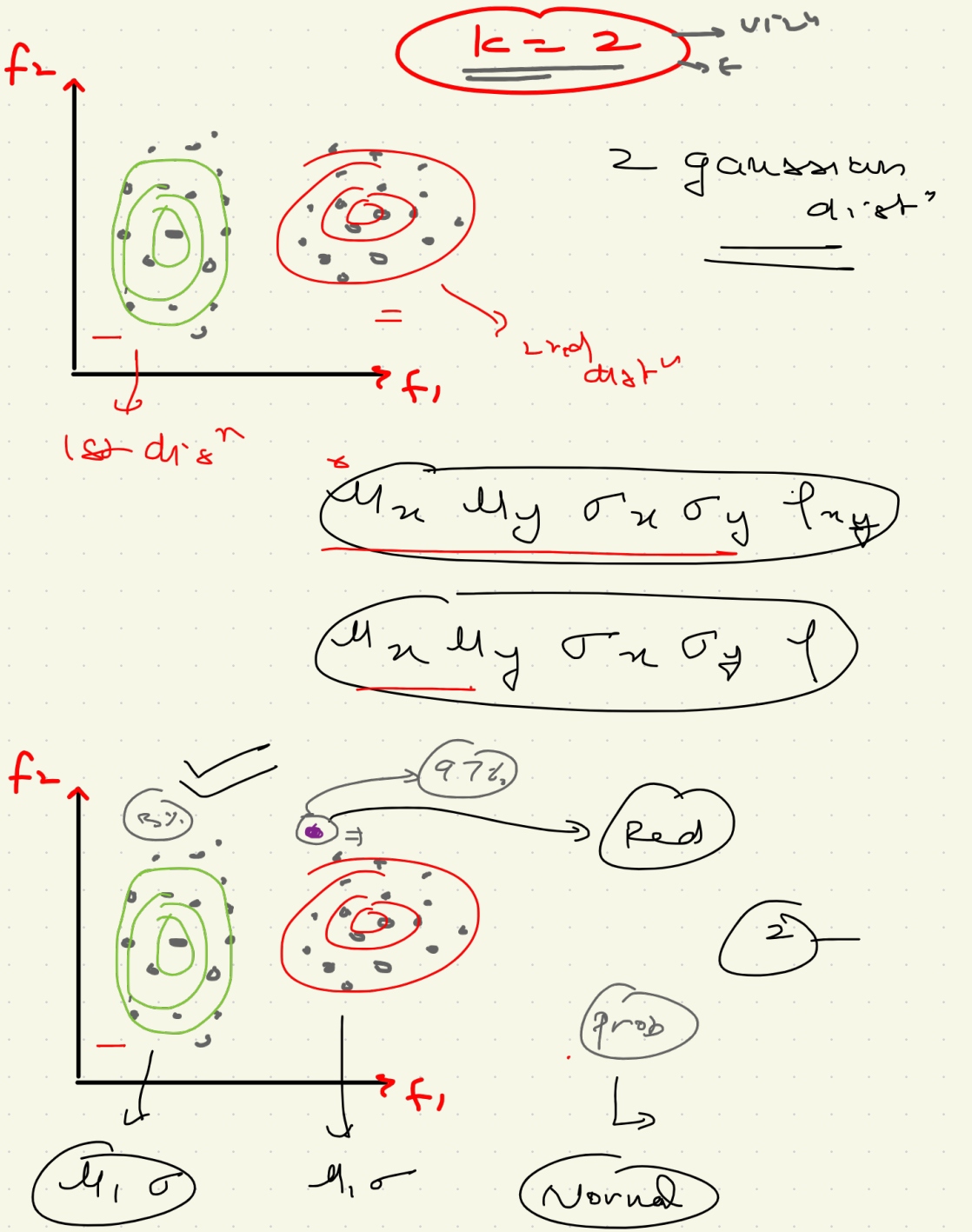
n peaks → multimodal

1 peak → Unimodal



μ, σ μ, σ

GMM → CLUSTERING


k = 2
f2
f1
2 gaussian dist"
=
2 prob dist"
1st dist"
μx μy σx σy ρxy
μx μy σx σy ρ
97%
Red
2
prob
Normal
μ1 σ
μ1 σ

$$P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x_j - \mu)^2}{2\sigma^2}\right)$$

GMM

dist^n 1 dist^n

$$d \rightarrow \mu = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_d \end{bmatrix}$$

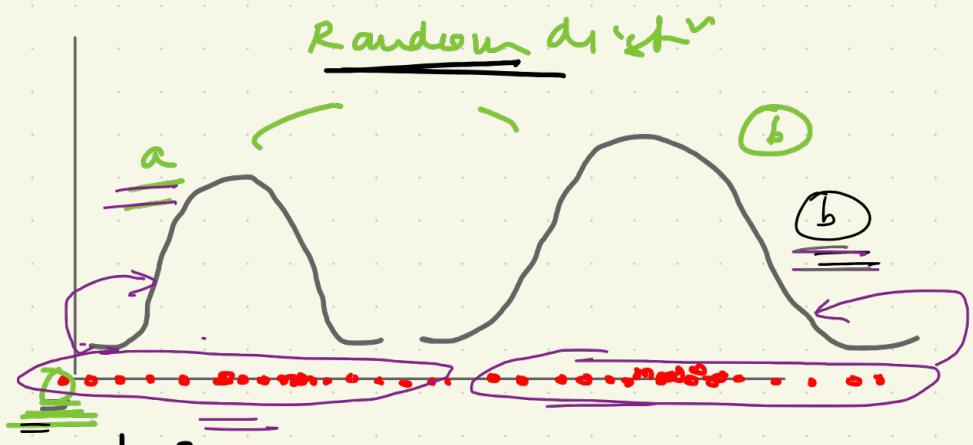
$$\Sigma = \begin{matrix} \\ p_1 \\ \\ f_d \end{matrix} \begin{bmatrix} f_1 & & f_d \\ \sigma_1^2 & & \\ & & \sigma_d^2 \end{bmatrix}$$

Expectation Maximization Algo [EM]



(1) initialization → Random params

$$X = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}^T$$ nDP



μ_1 → Random

$= x_{20}$

$\mu_2 = x_{33}$

$$\sigma_1 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$\sigma_2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

↳ (σ) Same

(2) Repeat until convergence

max-ilrn

$\mu_1, \mu_2, \sigma_1, \sigma_2, \phi$

DP

$\frac{0.1}{0.1+0.3}$ $\frac{0.3}{0.1+0.3}$

(a) E step

	$p(\mu_1)$	$p(\mu_2)$
x_1	0.1	0.3
x_2	0.2	0.04
x_3	0.2	0.4
.	.	.
.	.	.
x_n	.	.

Normalize →

	$p(\mu_1)$	$p(\mu_2)$
x_1	0.25	0.75
x_2	0.71	0.29
x_3	0.37	0.66
.	.	.
.	.	.
x_n	.	.

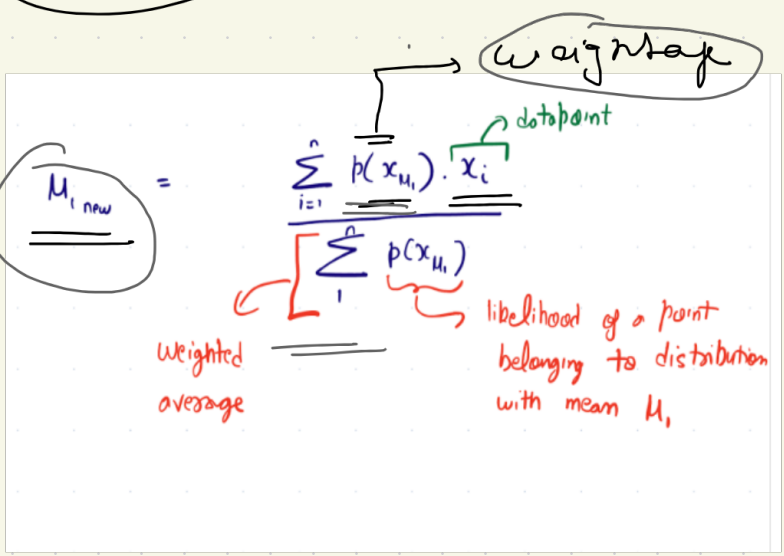
(4) Responsibilities

Assignment ⟹ ✗

μ_1 σ → updates

M step

σ, μ



$x_1 = 10, \quad x_2 = 20, \quad x_3 = 30$

(A) $\mu_{1\,new} = \frac{10 \times 0.25 + 20 \times 0.7 \times 30.33}{0.25 + 0.7 + 0.33}$

$\mu_{2\,new} =$

$\sigma_1^2 = \dfrac{\frac{1}{n}\sum p(\mu_{x_i}) \cdot (x - \mu_1)^2}{\sum p(\mu_{x_i})}$

σ_2

$$\mu_1 = \frac{1}{n}\sum_{i=1}^{n} P_{\mu_1}(x_i) \times x_i$$

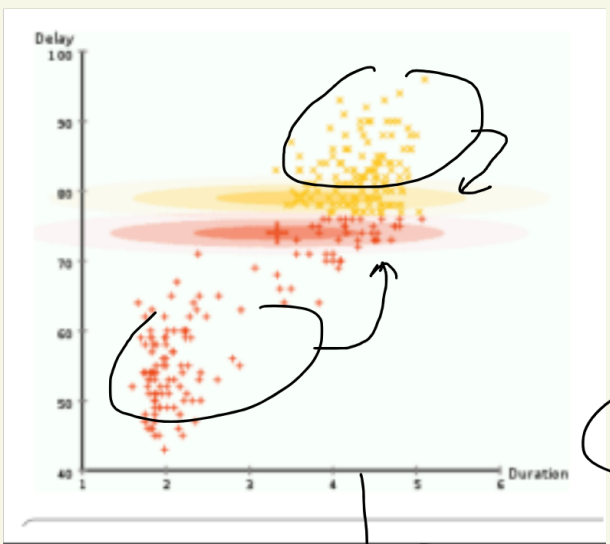
$\mu_2 =$

$\sigma_1^2 = \frac{1}{n}\sum_{i=1}^{n} P_{\mu_i}(x_i) \times (x_i - \mu_1)^2$

σ_2

$$\phi = \frac{1}{n} \sum_{i=1}^{n} p_{z_i}(x_i)(x - \mu_1)(x - \mu_2)$$

updating



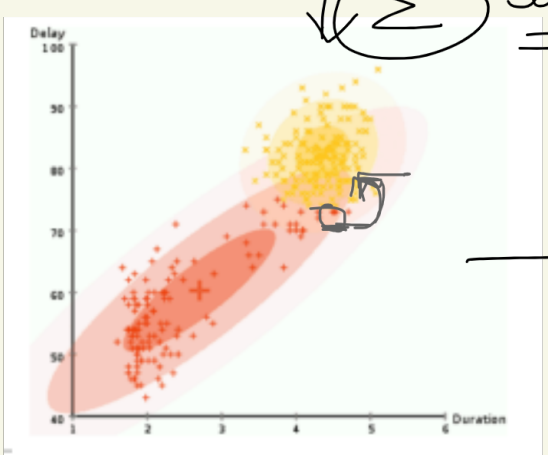
Step 1

2 distⁿ ⇒

σ, μ

Step 2

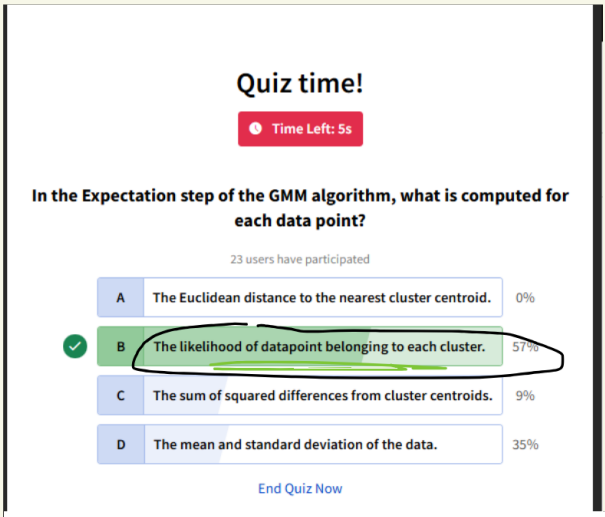
Σ μ = μ, σ

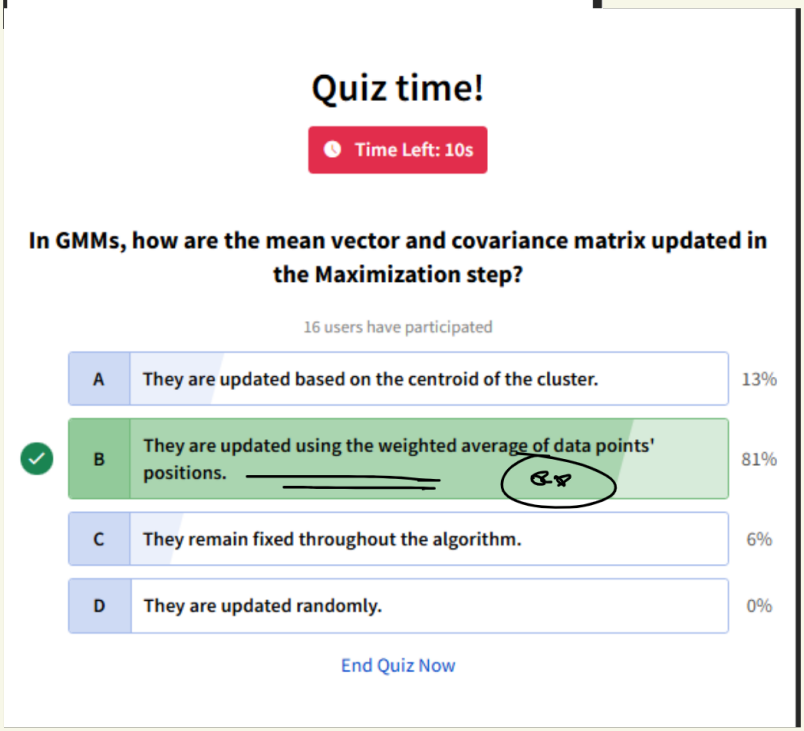


Responsibilities

μ_1 Σ

Stop


Quiz time!
Time Left: 5s
In the Expectation step of the GMM algorithm, what is computed for each data point?
23 users have participated
A
The Euclidean distance to the nearest cluster centroid.
0%
B
The likelihood of datapoint belonging to each cluster.
57%
C
The sum of squared differences from cluster centroids.
9%
D
The mean and standard deviation of the data.
35%
End Quiz Now


Quiz time!
Time Left: 10s
In GMMs, how are the mean vector and covariance matrix updated in the Maximization step?
16 users have participated
A
They are updated based on the centroid of the cluster.
13%
B
They are updated using the weighted average of data points' positions.
81%
C
They remain fixed throughout the algorithm.
6%
D
They are updated randomly.
0%
End Quiz Now

Quiz time!

Time Left: 13s

What is the primary difference between K-Means and Gaussian Mixture Models (GMMs)?

21 users have participated

A	K-Means requires specifying the number of clusters in advance, while GMMs do not.	19%
✓ B	K-Means assigns each point to the closest centroid with a hard assignment, while GMMs use soft assignment with probabilities.	76%
C	K-Means assumes that data is generated from a mixture of Gaussian distributions, while GMMs do not make this assumption.	5%
D	K-Means is more computationally efficient than GMMs.	0%

k-Means	GMM
Clusters are defined by their means	Clusters are defined by their means and their variance, modeled as Gaussians
Limitations if clusters are overlapping	Works if clusters are overlapping
Uses Euclidean distance to the mean	Uses the probability of X belonging to a cluster (generative)

